

FORMACIÓN DOCENTE 2026

Transformación Digital para la Docencia Técnica 3

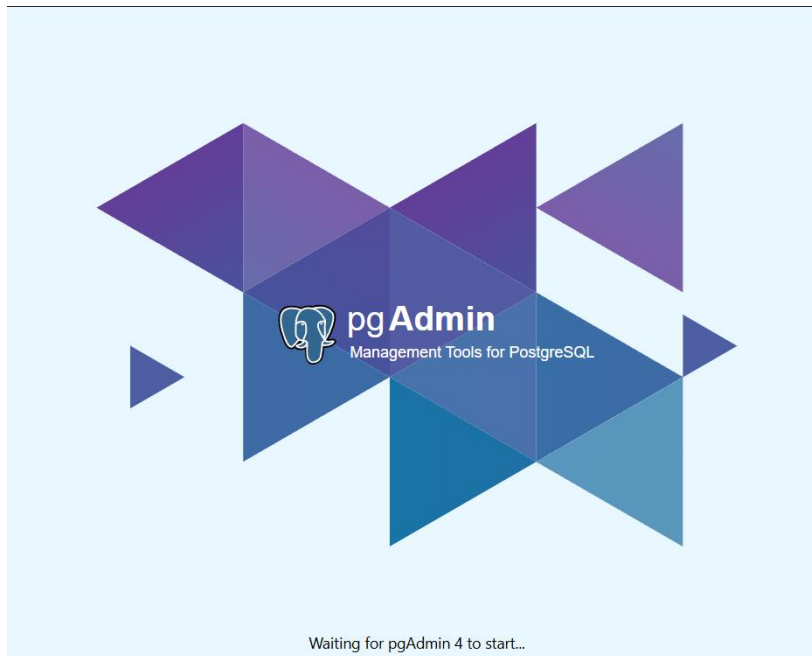
Módulo 5: Bases de Datos

Actividad de Evaluación: Script SQL con consultas CRUD y consultas JOIN funcionales sobre la base de datos existente

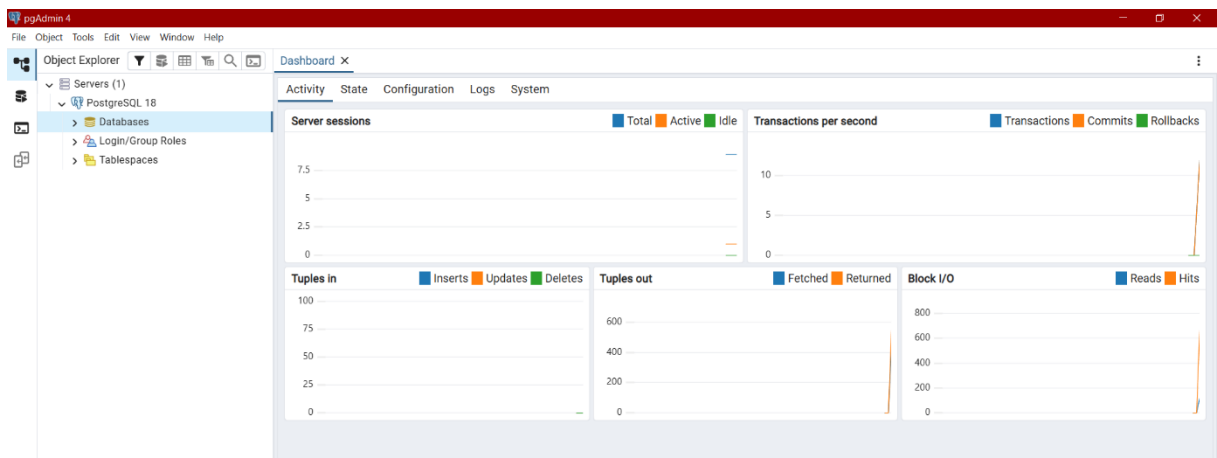
Estudiante: Roberto Abraham Méndez Clímaco

EVIDENCIAS

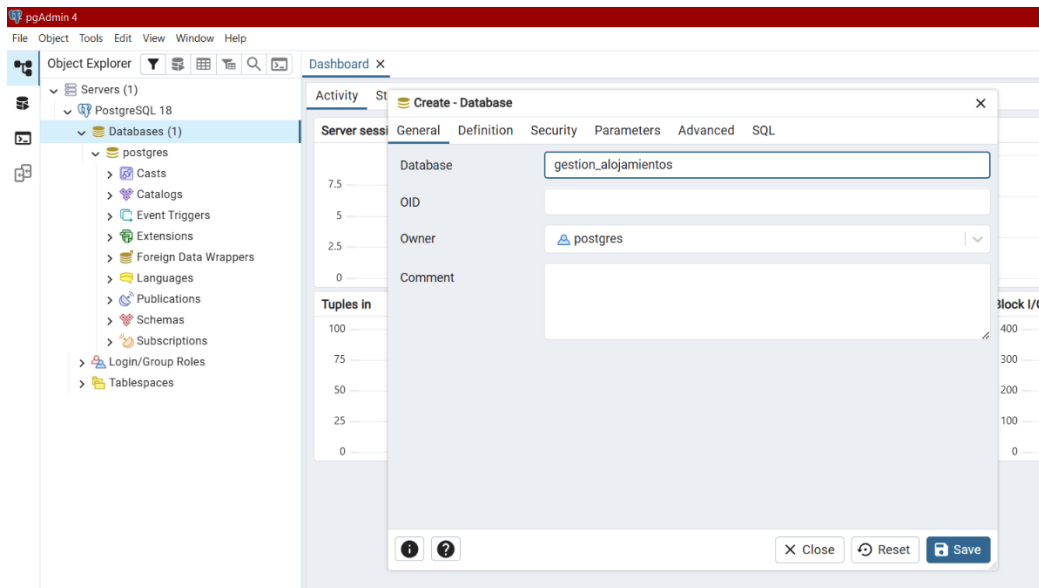
1. USO DE PGADMIN



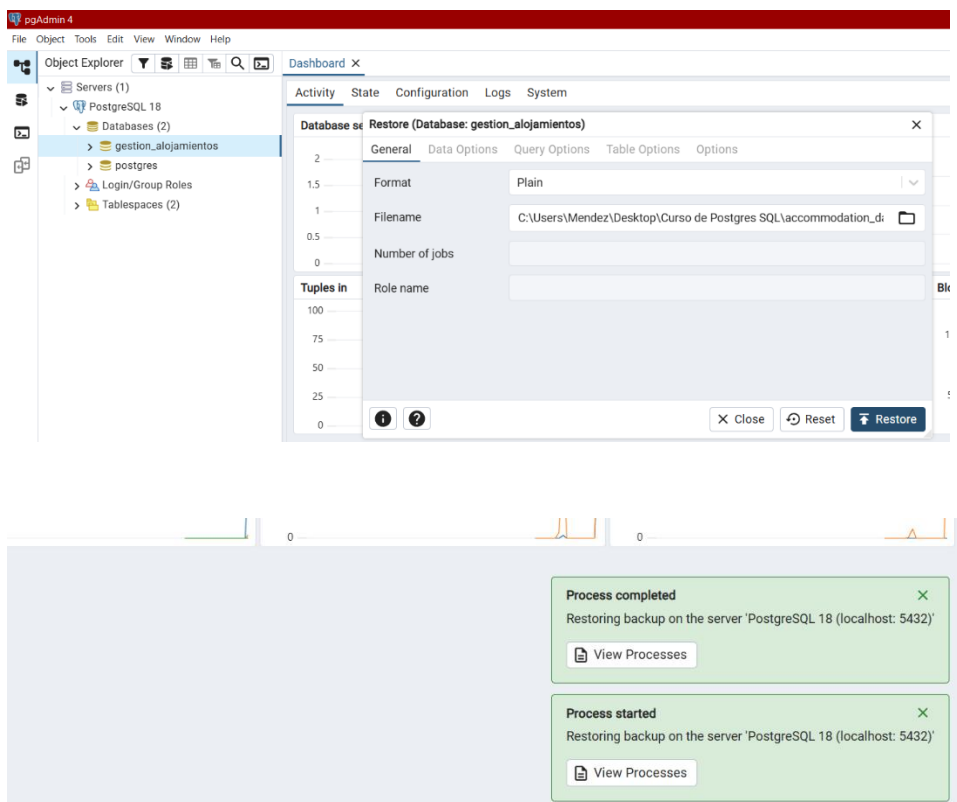
2. SERVIDOR LOCAL – OK

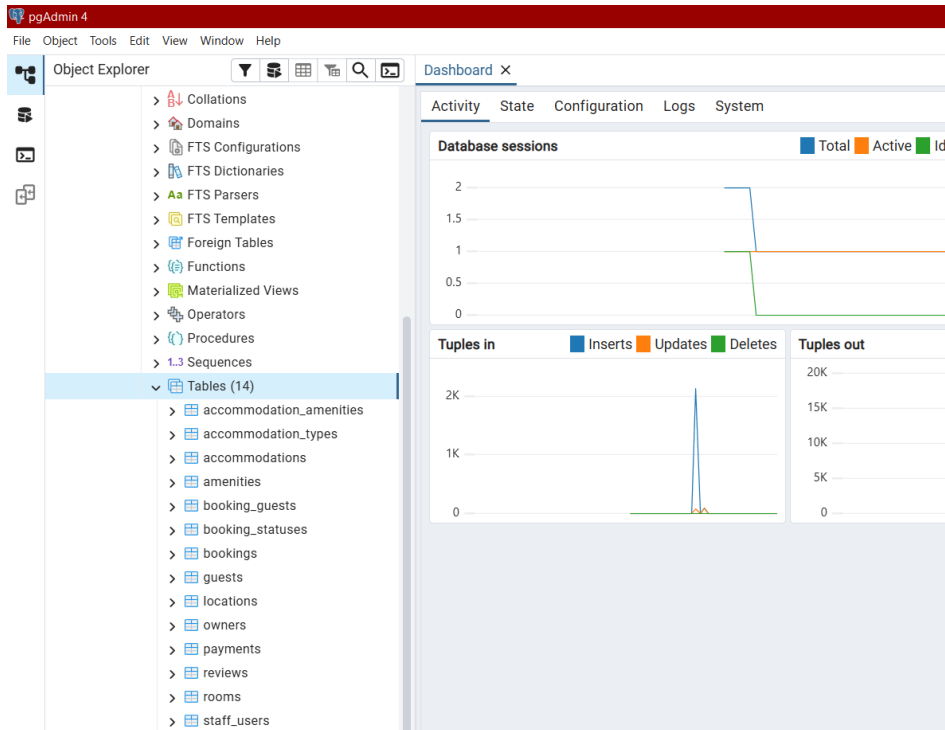


3. CREAR LA BASE DE DATOS LLAMADA: `gestion_alojamientos` PARA UTILIZAR EL BACK UP PROPORCIONADO



4. RESTAURACION DE LA BASE DE DATOS PROPORCIONADA





CONSULTAS SQL SOLICITADAS

1. Insertar propietario: Agregar un nuevo propietario

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is as follows:

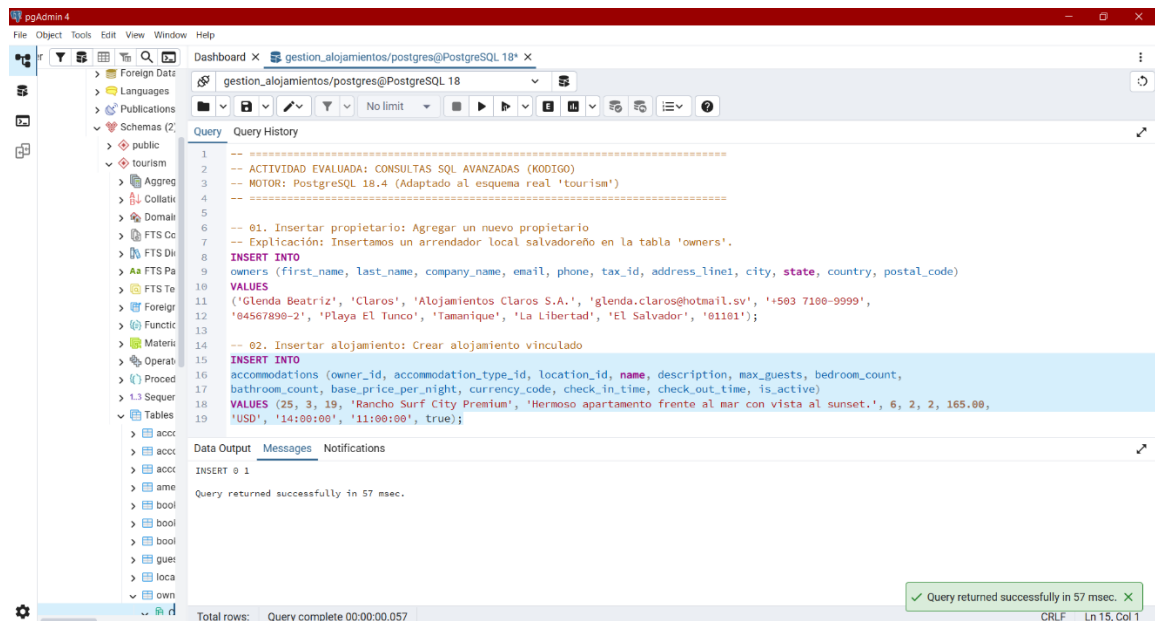
```
1 -- =====
2 -- ACTIVIDAD EVALUADA: CONSULTAS SQL AVANZADAS (CODIGO)
3 -- MOTOR: PostgreSQL 18.4 (Adaptado al esquema real 'tourism')
4 -- =====
5
6 -- 01. Insertar propietario: Agregar un nuevo propietario
7 -- Explicación: Insertamos un arrendador local salvadoreño en la tabla 'owners'.
8 INSERT INTO
9 owners (first_name, last_name, company_name, email, phone, tax_id, address_line1, city, state, country, postal_code)
10 VALUES
11 ('Glenda Beatriz', 'Claros', 'Alojamientos Claros S.A.', 'glenda.claros@hotmail.sv', '+503 7100-9999',
12 '04567890-2', 'Playa El Tunco', 'Tamanique', 'La Libertad', 'El Salvador', '01101');
13
14
```

The 'Data Output' tab shows the result of the query:

```
INSERT 0 1
Query returned successfully in 67 msec.
```

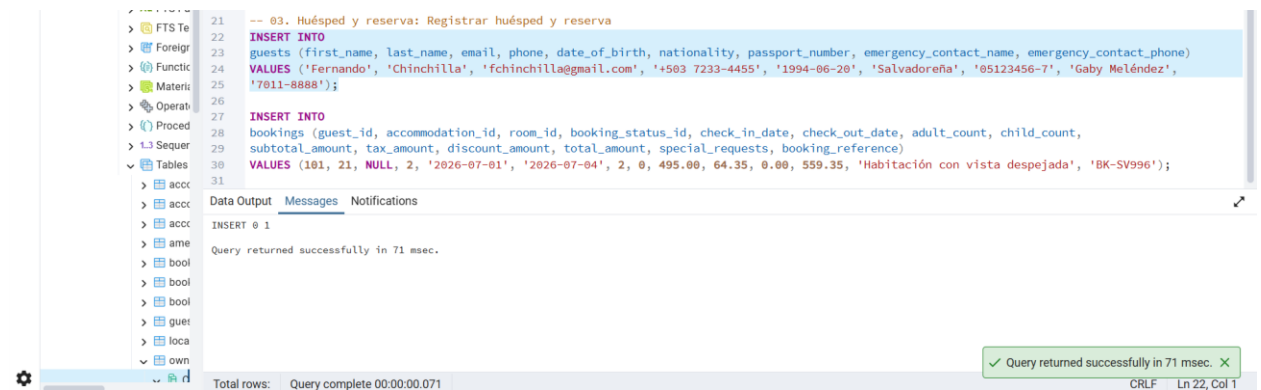
A green notification box at the bottom right states: 'Query returned successfully in 67 msec. X'

2. Insertar alojamiento: Crear alojamiento vinculado

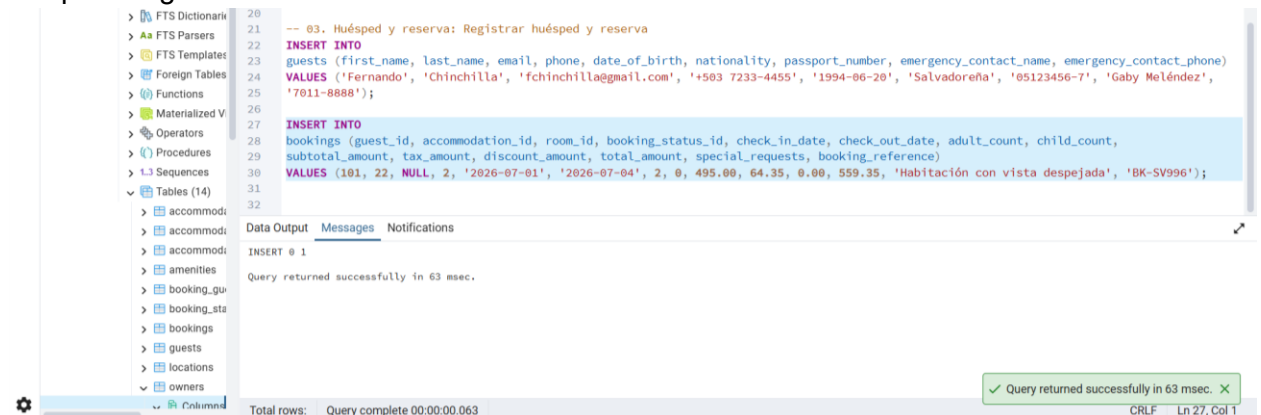


3. Huésped y reserva: Registrar huésped y reserva

Primero registro al cliente



Después registro la reserva



4. Insertar pago: Registrar pago

```
-- 04. Insertar pago: Registrar pago
-- Explicación: Registramos el cobro de la reserva en la tabla 'payments' usando Chivo Wallet como método de pago local.
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_reference, notes)
VALUES (103, 559.35, 'Chivo Wallet', 'Completed', 'TX-CHIVO-99827', 'Pago completo de estadia en Surf City');
```

Query returned successfully in 48 msec.

Total rows: Query complete 00:00:00.048

5. Alojamientos activos: Filtrar activos

```
-- 05. Alojamientos activos: Filtrar activos
SELECT * FROM accommodations
WHERE is_active = true;
```

accommodation_id [PK] bigint	owner_id bigint	accommodation_type_id integer	location_id bigint	name character varying (150)	description text
1	1	9	7	Rustic Lodge Ville	Asimismo mes principios peso joven.
2	2	14	2	Rustic Hideaway Ville	Democratic change vote participant institution.
3	3	10	7	Boutique Retreat Ville	Eran buenos natural. General victoria rey su.
4	5	10	4	Panoramic Stay Ville	Quinze exiner auquel déposer personne Rejoindre

Successfully run. Total query run

Total rows: 4 Query complete 00:00:00.006

6. Huéspedes por país: Filtrar por nacionalidad

```
-- 06. Huéspedes por país: Filtrar por nacionalidad
SELECT * FROM guests
WHERE nationality = 'Honduras' OR nationality = 'Salvadoreña';
```

guest_id [PK] bigint	first_name character varying (100)	last_name character varying (100)	email character varying (150)	phone character varying (30)	date_of_birth date	nationality character varying (100)
1	Débora	Holmes	masivan@example.org	+34724888992	1947-11-07	Honduras
2	Fernando	Chinchilla	fchinchilla@gmail.com	+503 7233-4455	1994-06-20	Salvadoreña

Successfully run. Total query run

Total rows: 2 Query complete 00:00:00.105

7. Reservas por fechas: Uso de BETWEEN

```
-- 07. Reservas por fechas: Uso de BETWEEN
-- Buscamos las reservas que se hicieron para entrar en el rango de junio de 2025
SELECT * FROM bookings
WHERE check_in_date BETWEEN '2025-06-01' AND '2025-06-30';
```

booking_id [PK] bigint	guest_id bigint	accommodation_id bigint	room_id bigint	booking_status_id integer	check_in_date date	check_out_date date	adult_count integer	child_count integer	total integer
1	1	39	15	4	2025-06-11	2025-06-23	4	3	
2	2	77	17	5	2025-06-08	2025-06-16	4	0	
3	3	41	9	20	2025-06-11	2025-06-25	4	0	
4	35	93	11	2	2025-06-15	2025-06-20	4	3	

Successfully run. Total query run

Total rows: 4 Query complete 00:00:00.113

8. Actualizar precio: Modificar precio

```
54 -----
55 -- CATEGORÍA: UPDATE
56 -----
57
58 -- 08. Actualizar precio: Modificar precio
59 UPDATE accommodations
60 SET base_price_per_night = 195.00
61 WHERE accommodation_id = 25;
62
```

Data Output Messages Notifications

UPDATE 0

Query returned successfully in 98 msec.

✓ Query returned successfully in 98 msec. ✕

Total rows: Query complete 00:00:00.098 CRLF Ln 59, Col 1

9. Estado reserva: Actualizar estado

```
62
63 -- 09. Estado reserva: Actualizar estado
64 -- Cambio el estado de la reserva a CheckeIn (ID 3 de la tabla booking_statuses)
65 UPDATE bookings
66 SET booking_status_id = 3
67 WHERE booking_id = 103;
68
```

guests
locations
owners
Columns (15)
owner_id bigint
first_name character
last_name character
company_name character
email character
phone character
tax_id character
address_line1 character
address_line2 character
city character
state character
country character

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 59 msec.

✓ Query returned successfully in 59 msec. ✕

Total rows: Query complete 00:00:00.059 CRLF

10. Eliminar reseña: DELETE WHERE

```
70 -----
71 -- CATEGORÍA: DELETE (Consulta 10)
72 -----
73 -- 10. Eliminar reseña: DELETE WHERE
74 DELETE FROM reviews
75 WHERE review_id = 1;
76
```

bookings
guests
locations
owners
Columns (15)
owner_id bigint
first_name character
last_name character
company_name character
email character
phone character
tax_id character
address_line1 character
address_line2 character
city character
state character
country character

Data Output Messages Notifications

DELETE 1

Query returned successfully in 58 msec.

✓ Query returned successfully in 58 msec. ✕

Total rows: Query complete 00:00:00.058 CRLF Ln 74, Col 1

11. Reservas + huésped: INNER JOIN

booking_statuses
bookings
guests
locations
owners

Columns (15)
owner_id bigint
first_name character varying (100)
last_name character varying (100)
company_name character varying (100)
email character varying (255)
phone character varying (20)
tax_id character varying (10)
address_line1 character varying (255)
address_line2 character varying (255)
city character varying (100)
state character varying (100)
country character varying (100)

```
-- CATEGORÍA: JOIN (Consultas 11 a 13)

-- 11. Reservas + huésped: INNER JOIN
SELECT b.booking_id, b.booking_reference, g.first_name, g.last_name, b.check_in_date, b.total_amount
FROM bookings b
INNER JOIN guests g ON b.guest_id = g.guest_id;
```

Data Output Messages Notifications

Showing rows: 1 to 101 Page No: 1

	booking_id bigint	booking_reference character varying (50)	first_name character varying (100)	last_name character varying (100)	check_in_date date	total_amount numeric (10,2)
1	1	BK-NVQHW06X	Ian	Giner	2025-06-11	266.00
2	2	BK-6LFEWVZN	Karl-Josef	Louis	2025-06-08	2409.77
3	3	BK-5P50300L	Irmtraut	Hettner	2025-06-11	2000.69
4	4	BK-EC8UY057	Henri	Schmidtke	2025-05-27	1384.24
5	5	BK-D4N1U349	Claude	Miller	2025-07-31	

Total rows: 101 Query complete 00:00:00.081

Successfully run. Total query runtime:

12. Alojamiento completo: INNER JOIN múltiple

bookings
guests
locations
owners

Columns (15)
owner_id bigint
first_name character varying (100)
last_name character varying (100)
company_name character varying (100)
email character varying (255)
phone character varying (20)
tax_id character varying (10)
address_line1 character varying (255)
address_line2 character varying (255)
city character varying (100)
state character varying (100)
country character varying (100)

```
-- 12. Alojamiento completo: INNER JOIN múltiple
SELECT a.name AS alojamiento, o.first_name || ' ' || o.last_name AS propietario, g.first_name || ' ' || g.last_name AS huesped,
b.check_in_date
FROM bookings b
INNER JOIN accommodations a ON b.accommodation_id = a.accommodation_id
INNER JOIN owners o ON a.owner_id = o.owner_id
INNER JOIN guests g ON b.guest_id = g.guest_id;
```

Data Output Messages Notifications

Showing rows: 1 to 101 Page No: 1 of 1

	alojamiento character varying (150)	propietario text	huesped text	check_in_date date
1	Luxury Nest Ville	Cynthia Benavidez	Ian Giner	2025-06-11
2	Luxury Getaway los altos	Anunciación Real	Karl-Josef Louis	2025-06-08
3	Boutique Nest boeuf	Cynthia Benavidez	Irmtraut Hettner	2025-06-11
4	Rustic Hideaway Ville	Ismael Jacquot	Henri Schmidtke	2025-05-27
5	Rustic Retreat furt	Anunciación Real	Claude Miller	2025-07-31

Total rows: 101 Query complete 00:00:00.094

Successfully run. Total query runtime: 94 msec. 101 rows affected

13. Pagos + reservas: JOIN combinado

locations
owners

Columns (15)
owner_id bigint
first_name character varying (100)
last_name character varying (100)
company_name character varying (100)
email character varying (255)
phone character varying (20)
tax_id character varying (10)
address_line1 character varying (255)
address_line2 character varying (255)
city character varying (100)
state character varying (100)
country character varying (100)

```
-- 13. Pagos + reservas: JOIN combinado
SELECT p.payment_id, b.booking_reference, p.amount AS monto_pagado, p.payment_method, p.payment_status
FROM payments p
INNER JOIN bookings b ON p.booking_id = b.booking_id;
```

Data Output Messages Notifications

Showing rows: 1 to 91 Page No: 1

	payment_id bigint	booking_reference character varying (50)	monto_pagado numeric (10,2)	payment_method character varying (50)	payment_status character varying (50)
1	1	BK-NVQHW06X	2447.60	CreditCard	Completed
2	2	BK-6LFEWVZN	2437.43	CreditCard	Failed
3	3	BK-5P50300L	199.38	Cash	Refunded
4	4	BK-EC8UY057	136.97	Crypto	Refunded
5	5	BK-D4N1U349	602.10	BankTransfer	Refunded

Total rows: 91 Query complete 00:00:00.097

Successfully run. Total query runtime: 97 msec.

14. Sin reseñas: Incluye nulls

The screenshot shows a database query editor with a sidebar on the left containing a tree view of database objects: guests, locations, owners, and a list of columns (15) including owner_id, first_name, last_name, company_name, email, phone, tax_id, address_line1, address_line2, city, state, and country. The main editor displays a SQL query starting at line 106:

```
-- 14. Sin reseñas: Incluye nulls
SELECT a.accommodation_id, a.name, r.review_id, r.rating
FROM accommodations a
LEFT JOIN reviews r ON a.accommodation_id = r.accommodation_id
WHERE r.review_id IS NULL;
```

Below the query, the 'Data Output' tab is active, showing a table with 1 row and 4 columns: accommodation_id (bigint), name (character varying (150)), review_id (bigint), and rating (integer). The single row shows accommodation_id 22, name 'Rancho Surf City Premi...', review_id [null], and rating [null]. The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.077'. A green notification box on the right states 'Successfully run. Total query runtime: 77 msec. 1 rows affected'.

15. Sin reservas: Filtrar nulls

The screenshot shows the same database query editor. The SQL query starts at line 111:

```
-- 15. Sin reservas: Filtrar nulls
SELECT a.accommodation_id, a.name, a.base_price_per_night
FROM accommodations a
LEFT JOIN bookings b ON a.accommodation_id = b.accommodation_id
WHERE b.booking_id IS NULL;
```

The 'Data Output' tab shows a table with 0 rows and 3 columns: accommodation_id [PK] (bigint), name (character varying (150)), and base_price_per_night (numeric (10,2)). The status bar indicates 'Total rows: 0' and 'Query complete 00:00:00.101'. A green notification box on the right states 'Successfully run. Total query runtime: 101 msec. 0 rows affected'.

16. Total ingresos: SUM

The screenshot shows the same database query editor. The SQL query starts at line 118, with a category label 'CATEGORÍA: AGG - AGREGACIÓN' at line 119:

```
-- CATEGORÍA: AGG - AGREGACIÓN
-- 16. Total ingresos: SUM
SELECT SUM(amount) AS total_ingresos_globales
FROM payments
WHERE payment_status = 'Completed';
```

The 'Data Output' tab shows a table with 1 row and 1 column: total_ingresos_globales (numeric). The single row shows a value of 32152.53. The status bar indicates 'Total rows: 1' and 'Query complete 00:00:00.063'. A green notification box on the right states 'Successfully run. Total query runtime: 63 msec'.

17. Promedio rating: AVG

127 -- 17. Promedio rating: AVG
128 `SELECT AVG(rating) AS promedio_calificaciones FROM reviews;`
129
130

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	promedio_calificaciones
1	3.1864406779661017

Total rows: 1 Query complete 00:00:00.083

✓ Successfully run. Total query runtime: 83 msec. 1 row

18. Top alojamientos: COUNT + LIMIT

130 -- 18. Top alojamientos: COUNT + LIMIT
131 `SELECT accommodation_id, COUNT(booking_id) AS total_reservas`
132 `FROM bookings`
133 `GROUP BY accommodation_id`
134 `ORDER BY total_reservas DESC`
135 `LIMIT 1;`
136

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	accommodation_id	total_reservas
1	18	9

Total rows: 1 Query complete 00:00:00.098

✓ Successfully run. Total query runtime: 98 msec. 1 row

19. Más de 3 reservas: GROUP BY + HAVING

137 -----
138 -- CATEGORÍA: HAVING (Consulta 19)
139 -----
140
141 -- 19. Más de 3 reservas: GROUP BY + HAVING
142 `SELECT accommodation_id, COUNT(booking_id) AS total_reservas`
143 `FROM bookings`
144 `GROUP BY accommodation_id`
145 `HAVING COUNT(booking_id) >= 3;`
146

Data Output Messages Notifications

Showing rows: 1 to 20 Page No: 1 of 1

	accommodation_id	total_reservas
1	11	6
2	8	5
3	19	5
4	4	5
5	14	4

Total rows: 20 Query complete 00:00:00.117

✓ Successfully run. Total query runtime: 117 msec. 20 rows

20. Alojamiento más caro: Subquery

151 -- 20. Alojamiento más caro: Subquery
152 `SELECT name, base_price_per_night, currency_code`
153 `FROM accommodations`
154 `WHERE base_price_per_night = (SELECT MAX(base_price_per_night) FROM accommodations);`
155

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	name	base_price_per_night	currency_code
1	Panoramic Suite Ville	597.44	MXN

Total rows: 1 Query complete 00:00:00.095

✓ Successfully run. Total query runtime: 95 msec. 1 row

